Network Science

Lecture 05: Social Context and Link Formation

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From Structure to Context

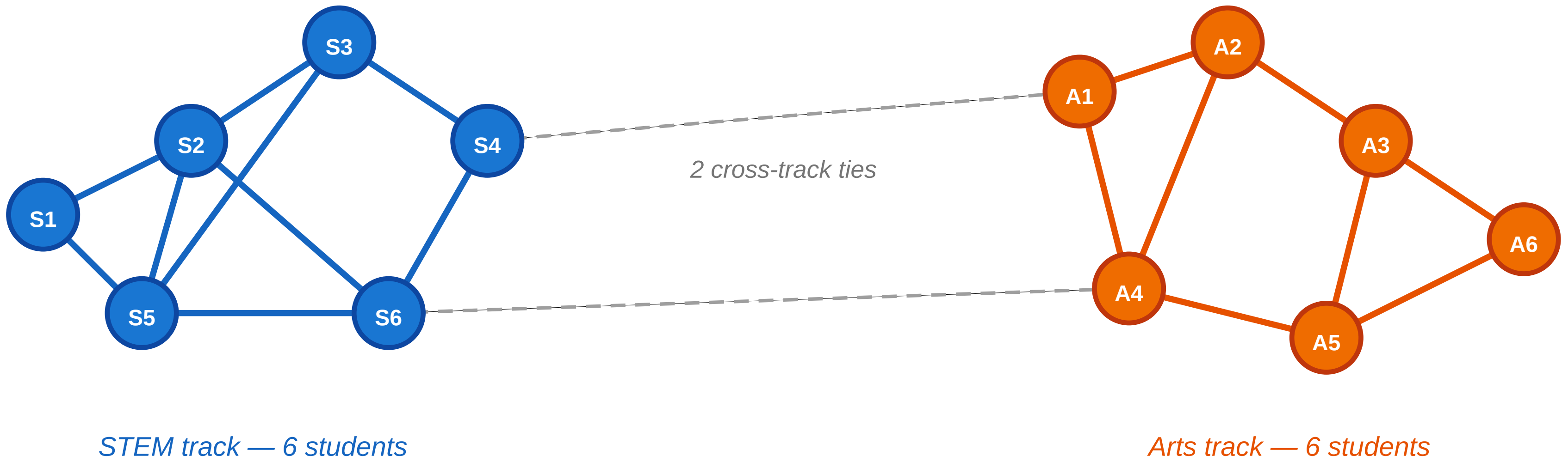
In Lectures 03 and 04 we classified edges (strong vs. weak) and groups (communities). The graph was colorless — every node was interchangeable. Today we ask: what happens when nodes carry *attributes* and edges carry *reasons*?

Consider a small classroom.

A Motivating Scenario

A small classroom: friendships and attributes

circles = students · color = track (blue = STEM, orange = Arts) · lines = friendships



Questions we'd like to answer from this picture

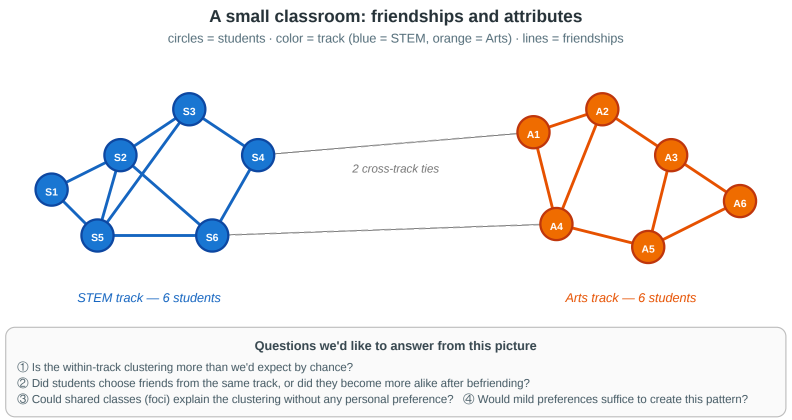
① Is the within-track clustering more than we'd expect by chance?
② Did students choose friends from the same track, or did they become more alike after befriending?
③ Could shared classes (foci) explain the clustering without any personal preference? ④ Would mild preferences suffice to create this pattern?

Twelve students, split equally between STEM (blue) and Arts (orange). Most friendships are within-track; only two cross-track ties exist. The clustering is visible — but why is it there?

Four Questions from One Picture

1. **Measurement.** Is the within-track clustering more than we would expect by chance — or could it happen under random mixing?
2. **Mechanism.** Did students choose friends from the same track (**selection**), or did friends influence each other to pick the same track (**socialization**)?
3. **Context.** Could shared classes or labs (**foci**) explain the clustering without any personal preference for similarity?
4. **Dynamics.** If every student has only a mild preference for same-track friends, could that suffice to produce this sharp segregation?

Every concept in today's lecture answers one of these.



A Different Kind of Gap

In L03–L04 the gap between theory and measurement was **computational**: MaxSTC and modularity maximization are NP-hard, so we used polynomial-time heuristics.

In L05 the gap is **causal**: even with infinite compute, a cross-sectional snapshot cannot tell us *why* a tie formed. The fix is not a better algorithm — it is richer data.

Question	What we want to know	What a snapshot shows	What we need
Homophily?	Is tie formation biased toward similarity?	Within-group edge fraction	Random-mixing baseline
Selection or socialization?	Which came first — similarity or the tie?	Both present simultaneously	Longitudinal data
Foci or preference?	Did shared context cause the tie?	Shared memberships and ties	Affiliation network model
Local → global?	Can mild preferences produce sharp segregation?	A segregated pattern	A generative model (Schelling)

Takeaway

> L03–L04 asked "what structure is there?" and hit a computational wall. L05 asks "why is the structure there?" and hits a causal wall. Network analysis requires both algorithmic tools and causal reasoning about the mechanisms behind the observed graph.

Homophily

> **Guiding Question:** Do people form ties because they are similar — and how would we know?

What Does Homophily Look Like?

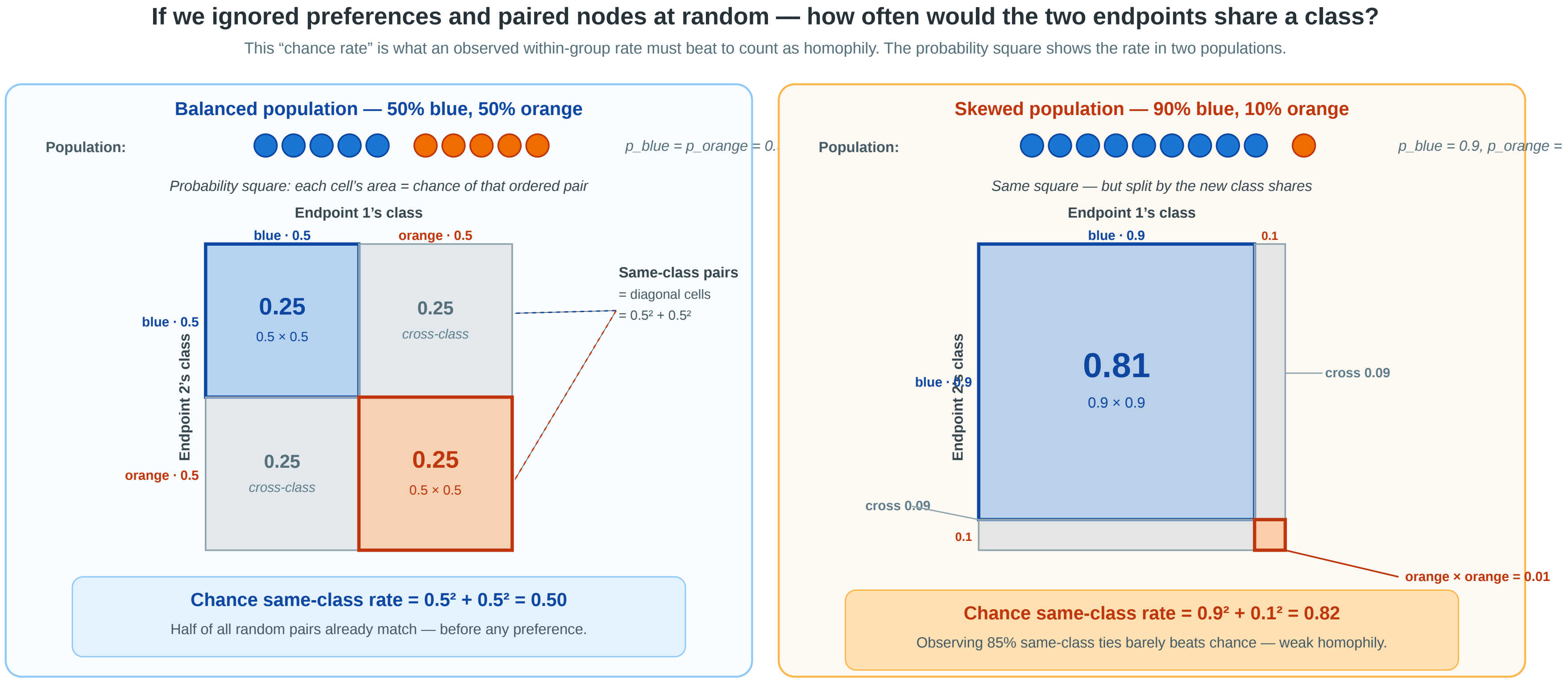
Homophily as similarity-biased tie formation

Solid edges are frequent within-group ties; dashed edges are rarer cross-group ties.

Homophily = tie formation is biased toward similarity. Within-group edges (colored) dominate; cross-group edges (gray) are rare. The pattern looks clear-cut — but a strong within-group share could also arise mechanically when one class is much larger than the other. The next slide makes that worry precise.

Homophily — What Counts as "More Than Chance"?

Before calling that pattern *homophily*, we have to ask: **how many same-class ties would we see if people paired up purely at random?** Only the *excess* over that chance rate is evidence of a preference for similarity.



Each cell's area is the chance of that ordered pair; the **colored diagonal cells** are the same-class outcomes. In the skewed population (right), 82% of random pairs already match — so an observed 85% within-group rate is almost noise. *Next slide:* we name this chance rate H_{base}.

Homophily — Formal Definition

Homophily. Edge formation is biased toward similarity in a node attribute x. Formally, for a categorical attribute with class shares $p_1, \ldots, p_k$:

- **Random-mixing baseline** (area of the diagonal cells in the probability square):

$$H_{\text{base}} = \sum_{i=1}^{k} p_i^2$$

- **Observed homophily:** the fraction of actual ties that connect same-class nodes, H_{obs}.
- **Homophily index:** how much the observed same-class rate exceeds the random baseline, normalized:

$$r = \frac{H_{\text{obs}} - H_{\text{base}}}{1 - H_{\text{base}}}$$

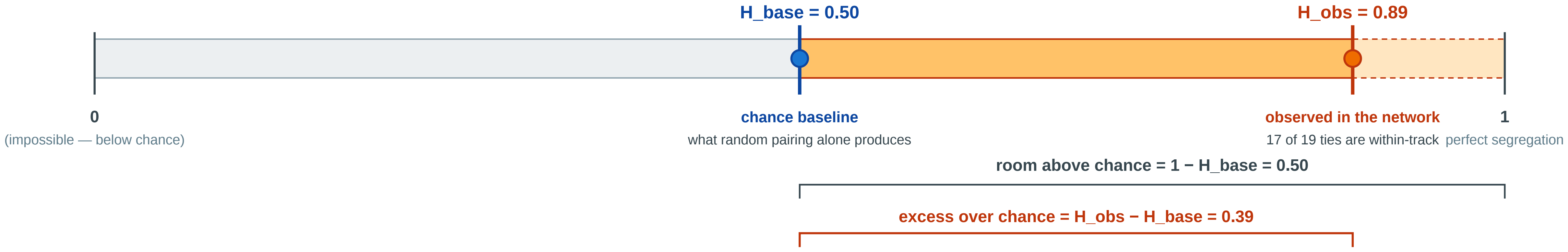
$r = 0$: random mixing. $r = 1$: perfect segregation. $r < 0$: heterophily (preference for *different* attributes).

Homophily — Three Numbers, One Story

Measure	What it tells you	Depends on
H_{base}	**The chance rate.** What random pairing alone would produce. Large when one class dominates the population.	Class shares only
H_{obs}	**What the data actually shows.** Count the within-class edges and divide by the total.	Observed edges only
r	**How far the network has moved from chance toward perfect segregation**, on a 0–1 scale. $r = 0$: chance. $r = 1$: perfect segregation. $r < 0$: cross-class preference (heterophily).	Both

How far has the network moved from chance toward perfect segregation?

The homophily index r is the fractional position of the observed rate on the “room above chance”.



r = excess ÷ room = 0.39 ÷ 0.50 = 0.78

The network has covered 78% of the way from chance mixing to perfect segregation.

The classroom example: $H_{\text{base}} = 0.50, H_{\text{obs}} = 0.89$. The observed rate has covered **78% of the available room** above chance — that is the homophily index r.

Homophily Baseline Trap

The **same observed within-group share** can mean very different things depending on the population composition. The table below holds $H_{\text{obs}} = 0.85$ fixed and varies the class shares:

Population shares	H_{base}	Room above chance	r	What you can honestly say
50% / 50%	0.50	0.50	0.70	Strong excess — 70% of the way to perfect segregation
70% / 30%	0.58	0.42	0.64	Substantial — 64% of available room used
90% / 10%	0.82	0.18	0.17	Barely above chance — almost nothing to explain
99% / 1%	0.98	0.02	< 0	*Below* chance — actually heterophilic!

Why this trap is so common. Journalists and researchers often report "X% of friendships are within-group" *without the baseline*. In a setting that is 80% one type and 20% another, H_{base} is already 0.68 — so an "82% within-group" headline is barely homophily. The trap appears whenever group sizes are imbalanced: school ethnicity studies, workplace gender networks, online platform demographics. The "Birds of a Feather" review (McPherson, Smith-Lovin & Cook 2001) insists on reporting effect sizes *relative to* a population-specific baseline for exactly this reason.

> Never interpret "X% of ties are within-group" without asking: **X% compared to what baseline?** The baseline depends on the population, not on what feels intuitively high.

Back to the Classroom

Both tracks have 6 students, so $p_{\text{STEM}} = p_{\text{Arts}} = 0.5$.

- $H_{\text{base}} = 0.5^2 + 0.5^2 = 0.50$
- 17 within-track edges of 19 total: $H_{\text{obs}} = 17/19 \approx 0.89$
- $r = (0.89 - 0.50)/(1 - 0.50) = 0.78$

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$r = 0.78$ is strong — far above random mixing. But it does not tell us *why*. Is it selection, socialization, or shared context?

Quick Check

In a school of 200 students, 60% are in track A and 40% in track B. Of all friendships, 78% are within-track.

1. Compute the random-mixing baseline H_{base}.
2. Compute the homophily index r.
3. Is r strong evidence of homophily? What could make it misleading?

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3. $r = 0.54$ is moderate-to-strong evidence of homophily. It could be misleading if:

- the two tracks occupy different buildings (shared context, not personal preference);
- students were *assigned* to tracks based on pre-existing friend groups (reverse causation);
- a third factor (e.g. family background) causes both track choice and friendship patterns (confounding).

Takeaway

Homophily is not just a visual pattern — it is a statistical claim that tie formation is biased toward similarity *relative to a random-mixing baseline*. The homophily index r quantifies the excess, but it cannot identify the mechanism.

Echo Chambers — The Hypothesis

Source: Cinelli et al. 2021

Claim	Online platforms expose users mostly to like-minded peers and sources.
Network signal	High homophily in the *interaction* network (likes, retweets, replies, comments) when nodes are coloured by political leaning.
Diffusion signal	Information circulates *within* groups rather than crossing them.

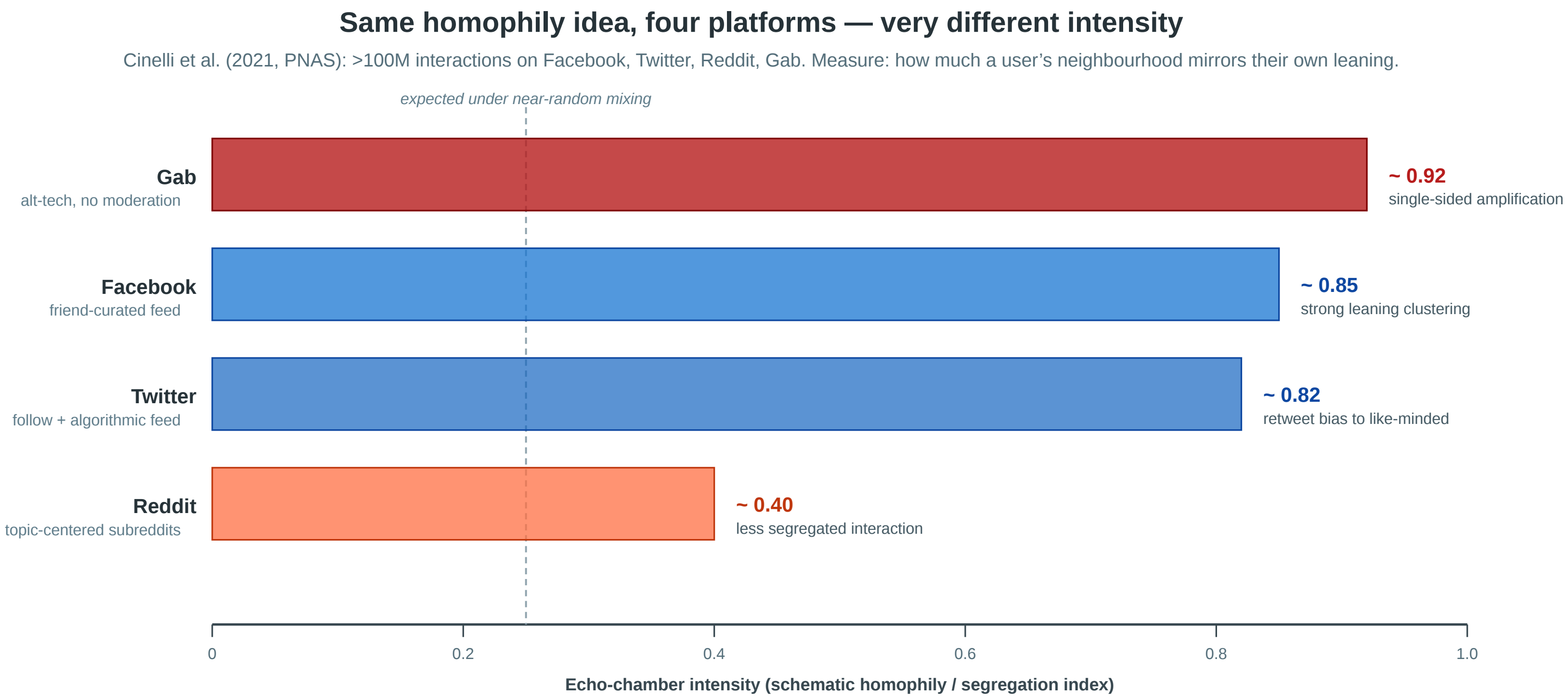
Why it's a homophily story. An echo chamber is not just "people who disagree online" — it is a region of the interaction graph where within-group ties dominate so heavily that the same content recirculates without leaving the cluster.

The empirical test (Cinelli et al. 2021, PNAS).

- Four platforms: **Facebook, Twitter, Reddit, Gab**.
- Over **100 million** interactions, hundreds of news outlets coded by political leaning.
- For each user, compute the *leaning gap* between the user and the people / outlets they interact with.
- A small leaning gap = the user lives in an echo chamber.

Different platform architectures (algorithmic feed, friend graph, subreddit topic, alt-tech) let us see whether echo chambers are universal or **architecture-dependent**.

Echo Chambers — Same Idea, Four Platforms

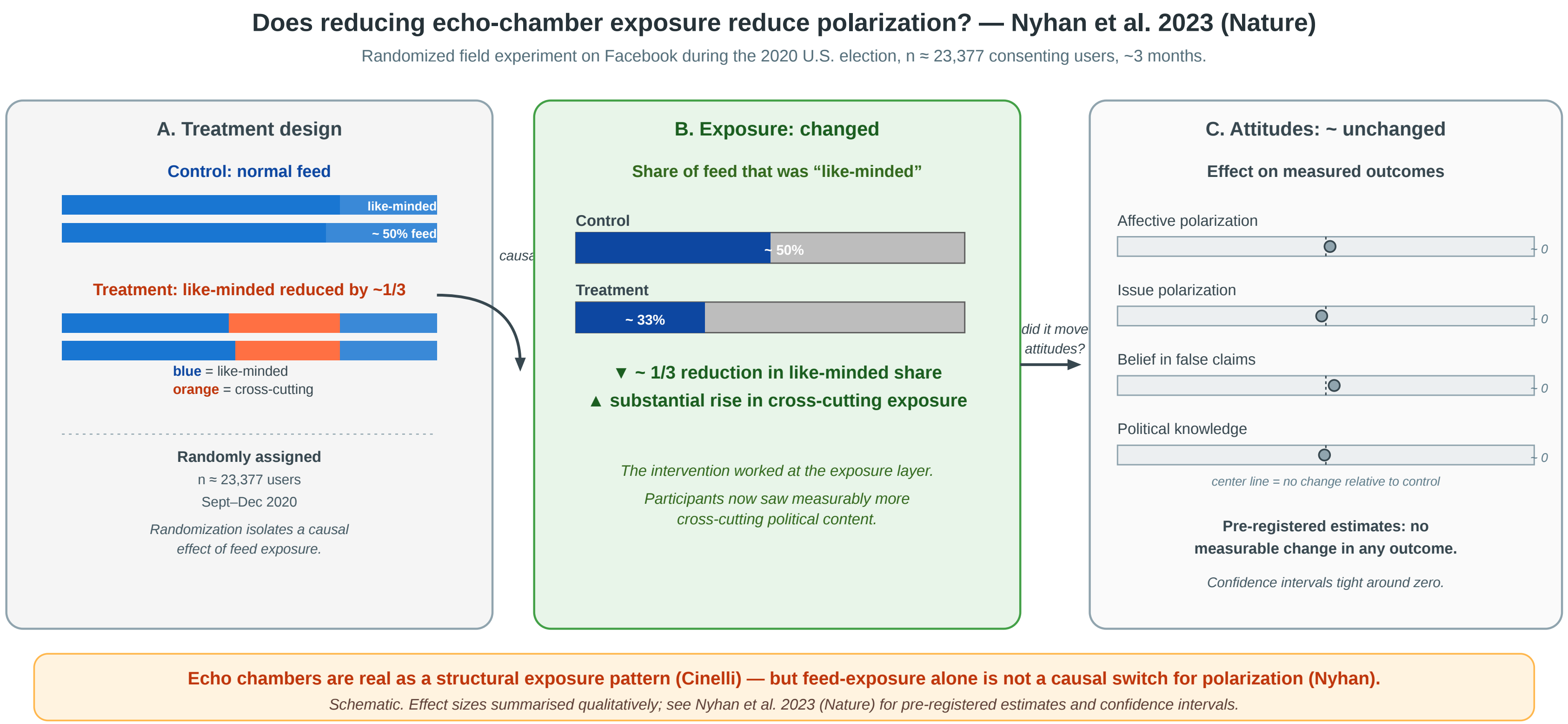


Schematic. Magnitudes are illustrative; the paper reports leaning-gap distributions, not a single index. The qualitative ranking (Gab > Facebook ≈ Twitter ≫ Reddit) matches the paper.

Architecture matters. Facebook and Twitter (friend / follow graphs amplified by ranking) show strong homophilic clustering by political leaning. **Gab** — small, unmoderated, single-sided audience — is even more extreme. **Reddit** is much less segregated: topic-centered subreddits force users to interact across leanings within the same subforum. The same homophily concept produces dramatically different intensities across designs.

Echo Chambers — Does Exposure Cause Polarization?

Source: Nyhan et al. 2023



Cinelli (2021) shows echo chambers *exist* as a structural pattern. **Nyhan et al. (2023, Nature)** ask whether that exposure *causes* polarization. In a pre-registered randomized field experiment around the 2020 U.S. election (n ≈ 23,377 consenting Facebook users), they reduced exposure to like-minded sources by about one-third for ~3 months. Cross-cutting exposure rose substantially — but **affective polarization, issue polarization, false beliefs, and political knowledge did not measurably change** (estimates tight around zero).

Structure ≠ causation. Dense homophily can *reinforce* information, but a few weeks of changed feed exposure may be too weak, too late, or too indirect to move stable political attitudes. The network concept ("echo chamber as homophilic region") is clearer than any causal claim about its effects.

Selection and Socialization

> **Guiding Question:** How can we distinguish whether similarity caused the tie or the tie caused the similarity?

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$$\text{homophily} = \text{selection} + \text{socialization} + \text{context}$$

Three Mechanisms, One Pattern

Selection. People choose to form ties with others who are *already* similar to them. Similarity precedes the tie.

Socialization (social influence). People become *more similar* to their contacts after the tie forms. The tie precedes similarity.

Contextual correlation. A shared environment (school, workplace, neighborhood) independently causes both the attribute and the tie. Neither causes the other directly.

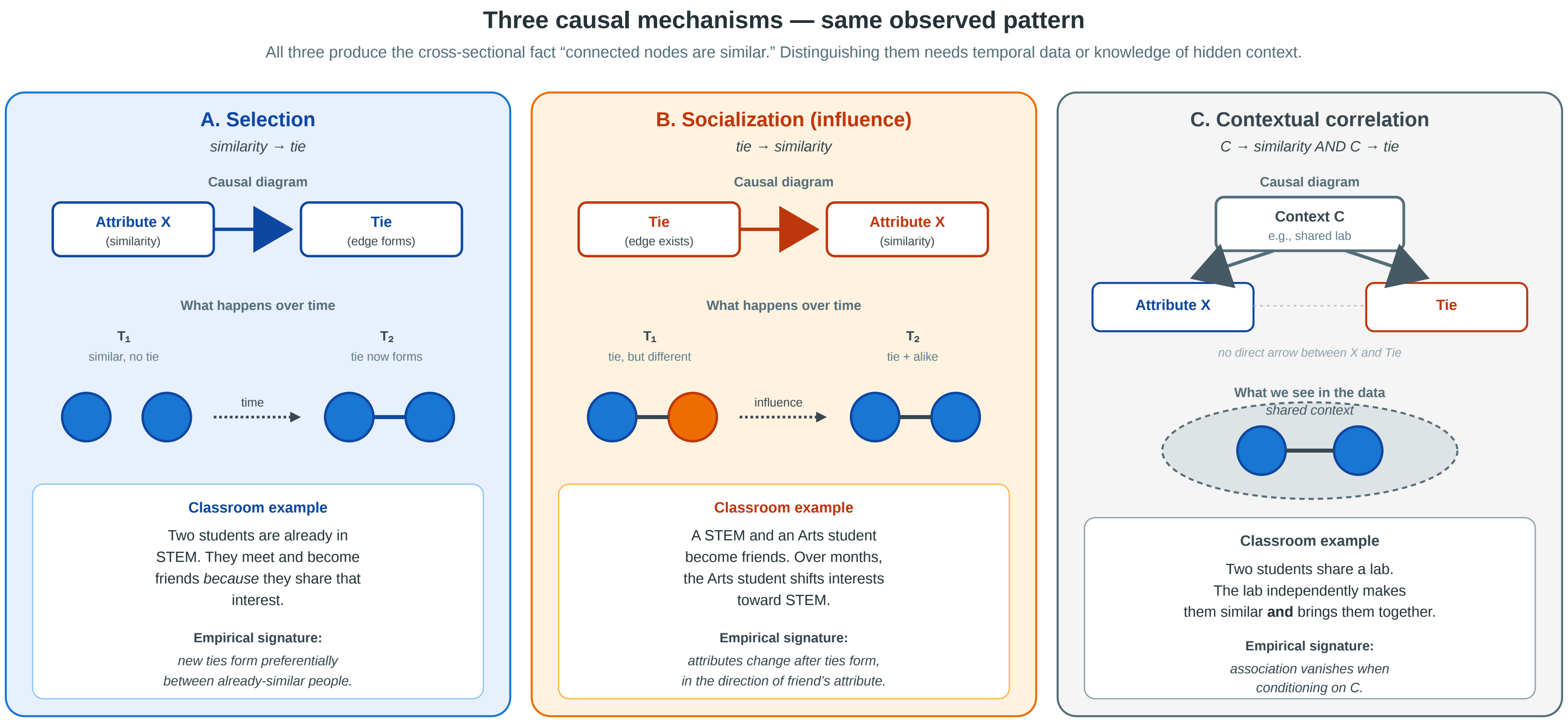
Same Snapshot, Three Stories

Suppose we observe: **STEM students mostly befriend STEM students.**

Mechanism	Concrete story	What data would help?
Selection	Students first choose STEM, then choose friends from the same track.	Friendships observed *after* track choice
Socialization	Friends influence each other to choose the same elective or track.	Track changes observed *after* friendships
Contextual correlation	STEM students share labs, schedules, and buildings, so they meet more often.	Timetables, room assignments, shared courses

The graph snapshot is identical in all three stories. The difference is the **time order** and the **unobserved context**.

Causal Diagrams



Empirical Test: Christakis & Fowler — Obesity in Social Networks

Christakis & Fowler (2007), *The Spread of Obesity in a Large Social Network Over 32 Years*, New England Journal of Medicine 357, 370–379.

They analyzed the **Framingham Heart Study** social network — 12 067 people tracked over **32 years** with repeated measurements of both friendship ties and body mass index. This made it possible to observe the *temporal sequence* of tie formation and weight change.

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Headline result. A person's probability of becoming obese increased by **57%** if a friend became obese in the same interval. The effect was stronger for mutual friends and showed **directionality** — it operated in the direction of the friendship nomination, suggesting socialization (influence) rather than pure selection.

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What the Study Could Not See

1. Confounders. Cohen-Cole & Fletcher (2008) and Lyons (2011) showed that the same statistical method would "detect" social contagion of **height** and **acne** — traits that clearly do not spread through social influence. The method may confuse *shared environment* with *social influence* when the confounders evolve over time.

2. Homophily in tie formation. The study adjusted for *existing* attribute similarity but could not fully exclude the possibility that people selectively *formed* or *maintained* ties with others on similar weight trajectories — a dynamic form of selection invisible to static controls.

3. Missing network. The Framingham data records nominated friends, not the full social network. Influence may flow through unobserved ties (coworkers, family, neighbors) that correlate with both friendship nominations and weight change.

Quick Check

You observe that smokers tend to befriend other smokers in a college dorm.

1. What kind of data would you need beyond a single snapshot to distinguish selection from socialization?
2. Sketch one observation pattern that would favor *selection* and one that would favor *socialization*.
3. Even with longitudinal data, what kind of confound would remain hard to rule out?

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3. Shared *environment* — e.g. dorm floor placement causing both friendship and smoking exposure simultaneously. Even with temporal data, confounders that evolve over time alongside both the network and the attribute are hard to rule out without randomization.

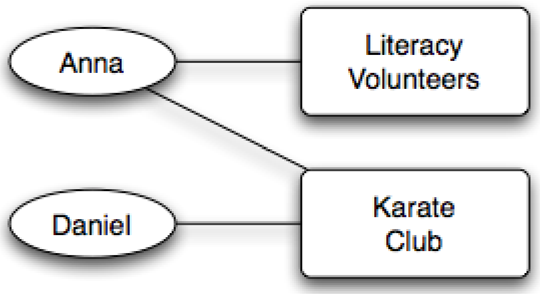
Takeaway

Cross-sectional data cannot distinguish selection from socialization — the same homophily index r is consistent with both. Longitudinal network data is far more informative, but even multi-decade panel studies cannot fully eliminate confounding by evolving shared environments.

Affiliation Networks

> **Guiding Question:** How do we represent the social settings — classes, clubs, workplaces — in which ties form?

Bipartite Affiliation Model



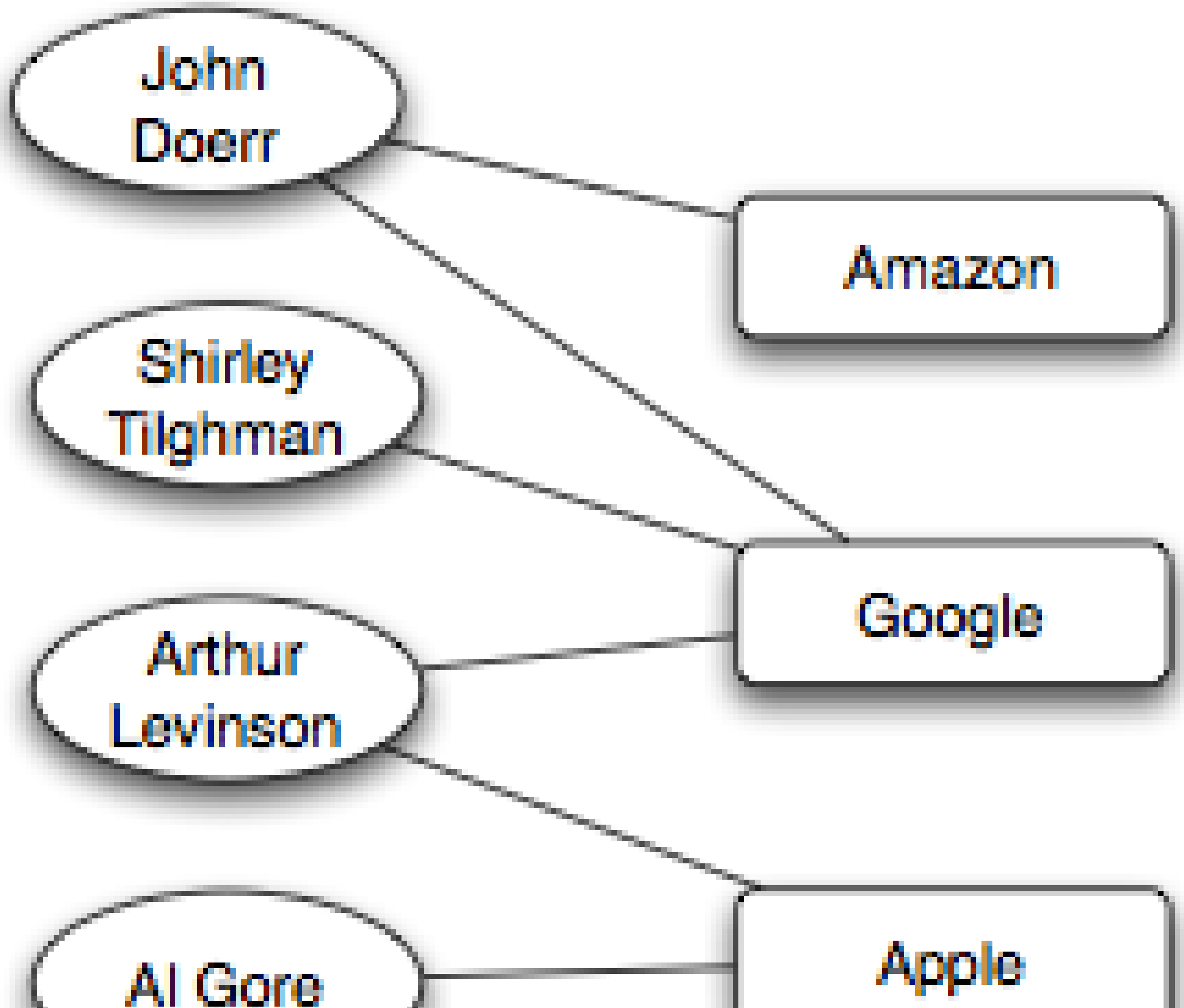
Source: Easley & Kleinberg 2010

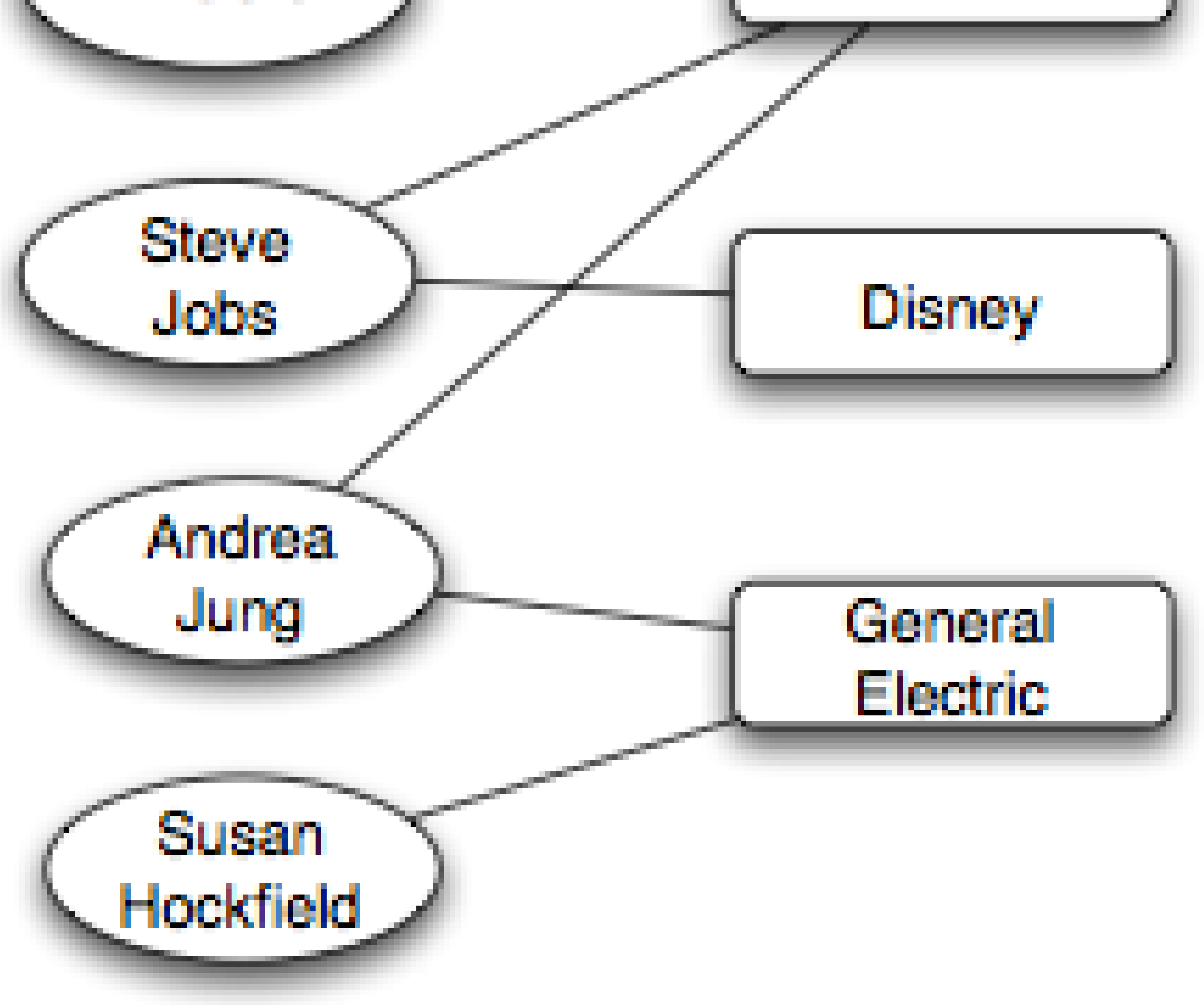
An **affiliation network** is a bipartite graph $G = (P \cup F, E)$:

- P: actor nodes (persons)
- F: focus nodes (groups, locations, classes, organizations)
- E: edges only between P and F — "person p participates in focus f"

Two persons sharing a focus creates a *potential* for a person–person tie.

Example — Corporate Board Interlocks


John Doerr
Shirley Tilghman
Arthur Levinson
Al Gore
Amazon
Google
Apple



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Reading the Board Network

An affiliation network is bipartite, but we usually want to ask one-sided questions about *people* or *firms*. The trick is to **project** the bipartite graph onto one side of the partition. There are two such projections, each answering a different question.

Projection	Nodes	Edge	Edge weight	Question it answers
Director network *(person side)*	People	Two directors sit on ≥ 1 shared board	Number of shared boards	Who can exchange career info / governance norms? Who is structurally similar?
Company network *(focus side)*	Firms	Two firms share ≥ 1 director	Number of shared directors	Where can strategy or practices diffuse? Which firms are strategically interlocked?

Toy example. Three directors d_1, d_2, d_3 on two boards b_1, b_2; $d_1, d_2 \in b_1$ and $d_2, d_3 \in b_2$.

- *Director projection:* $d_1!-!d_2$, $d_2!-!d_3 - d_2$ is a bridge.
- *Company projection:* $b_1!-!b_2$ — one edge, via the shared director d_2.

The projection **destroys information**: once projected, a $d_1!-!d_2$ edge no longer tells you *which* board mediated it. You cannot recover the bipartite graph from either one-mode projection alone.

Three rules of thumb.

1. **A projection edge = opportunity for exposure**, not a friendship and not a confirmed information channel. A quarterly board meeting is a weak channel.
2. **Density rises sharply.** A board of size s contributes $\binom{s}{2}$ projected edges from a single focus — a 12-person board adds 66 director-director edges. Projections look denser than the underlying network.
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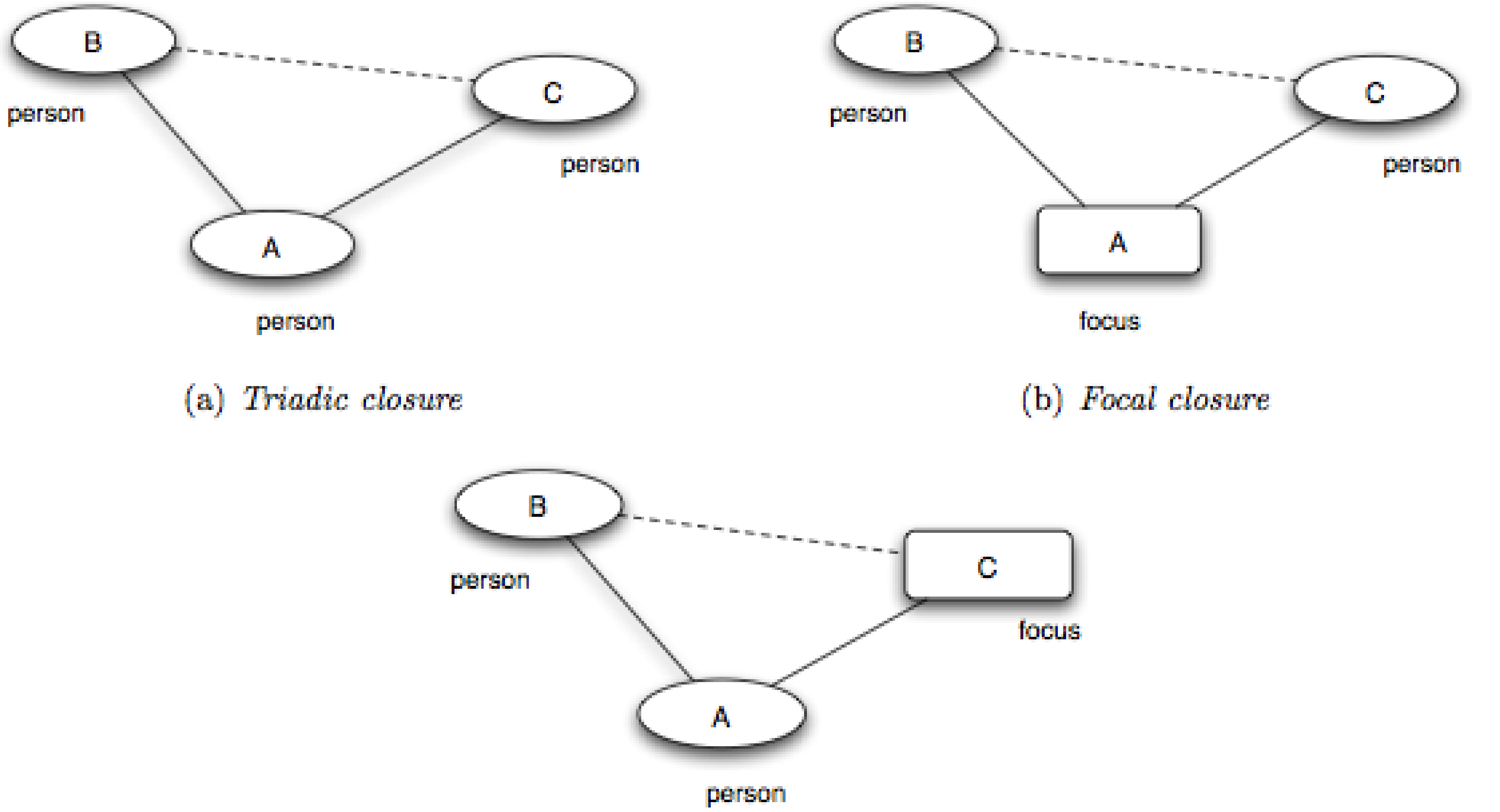
Reading rule. A projection edge means *"shared context exists"* — nothing more. Friendship, influence, and information flow all need additional evidence on

Three Closure Processes



Source: Easley & Kleinberg 2010

Triadic closure. Friend-of-friend becomes friend — a person–person–person path closes into a triangle. *(Same mechanism as Lecture 03.)*

Focal closure. Two persons sharing a focus become friends — a person–focus–person path closes into a person–person tie. *(New — the focus mediates the introduction.)*

Membership closure. A friend introduces you to a focus you did not belong to — a person–person–focus path closes into a person–focus tie. *(New — the friend mediates membership.)*

Closure Processes — Classroom Examples

Closure type	Open path	What closes?	Classroom example
Triadic	student → friend → student	A new friendship	"My lab partner introduces me to their friend."
Focal	student → elective → student	A new friendship	"We sit in the same statistics elective and start talking."
Membership	student → friend → club	A new membership	"My friend invites me to join the robotics club."

Only triadic closure needs a shared friend. Focal closure can create ties among strangers through repeated exposure in the same setting.

Quick Check

For each scenario, identify whether triadic, focal, or membership closure is at work.

1. Two students take the same elective and become friends.
2. A new graduate student joins the same lab as her advisor's other student, who introduces her.
3. A friend invites you to their book club, and you join.

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2. Triadic closure — the advisor is the shared friend that closes the triangle.

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3. Membership closure — your friend brings you into a focus you did not previously belong to.

Takeaway

Affiliation networks make social context explicit. Focal closure and membership closure are distinct tie-formation mechanisms that can produce homophily-like patterns without any personal preference for similarity — making them a critical alternative explanation in any homophily analysis.

Online Link Formation

Guiding Question: What can large-scale online data teach us about how ties form over time?

The Setting — Kossinets & Watts (2006)

Data. A large U.S. research university's internal email logs, ~~1 year, **~~43,000 students/faculty/staff**, **~14 million** messages — plus full course-enrollment records and organizational-affiliation rosters.
Why it matters. For the first time, all three closure types could be observed *with timestamps* on a single population: who emails whom, who shares which classes, who joins which clubs.

What was measured. At every time step, for each pair (u, v) *not yet connected*, count the number k of "closing paths": shared friends (triadic), shared foci (focal), or friend-mediated foci (membership). Then look at the probability that the pair connects in the next interval.
The output is a curve: $P(\mathrm{new\ tie});;\ \mathrm{vs.};;\ k$.

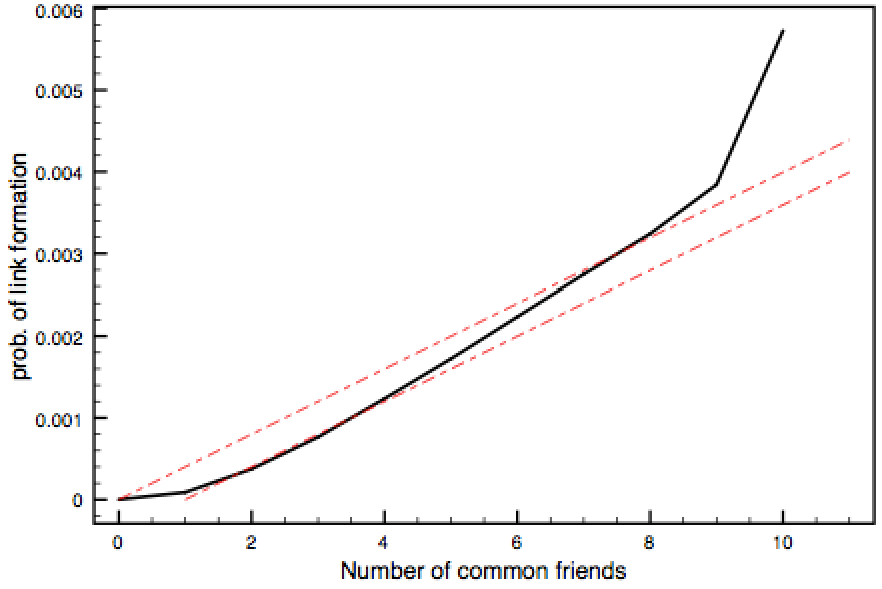
The Setting — Kossinets & Watts (2006)

Data. A large U.S. research university's internal email logs, ~~1 year, **~~43,000 students/faculty/staff**, **~14 million** messages — plus full course-enrollment records and organizational-affiliation rosters.
Why it matters. For the first time, all three closure types could be observed *with timestamps* on a single population: who emails whom, who shares which classes, who joins which clubs.

What was measured. At every time step, for each pair (u, v) *not yet connected*, count the number k of "closing paths": shared friends (triadic), shared foci (focal), or friend-mediated foci (membership). Then look at the probability that the pair connects in the next interval.
The output is a curve: $P(\text{new tie});;\text{vs.};;k$.

This is the dataset behind the three closure-curve figures in Easley & Kleinberg Ch. 4. The timestamped data finally separates *"what existed at t"* from *"what formed by $t + 1$"* — the missing ingredient in homophily snapshots.

Triadic Closure at Scale



Source: Easley & Kleinberg 2010

Setup. Two people who have **never emailed each other** but share k mutual email contacts. What is the probability they exchange email within the next interval?

k shared friends	$P(\text{new tie})$
0	very low (baseline)
1	sharply higher
2	higher still
5+	curve flattens — saturation

Two readings.

1. **Mechanism.** Mutual friends create *opportunities*: introductions, shared

interests, social pressure to know each other.

2. **Diminishing returns.** The 1st shared friend matters far more than the 10th — mutual friends are *not* independent channels; they often belong to the same group.

A Toy Model for the Shape

If each shared contact independently offers an introduction opportunity with probability q, then the probability of closure given k shared contacts is:

$$P(\text{new tie} \mid k \text{ shared}) = 1 - (1 - q)^k$$

This *saturating* curve rises steeply at first and flattens as k grows — matching the empirical shape qualitatively. But real curves often flatten *faster* than this model predicts, suggesting that shared contacts are not fully independent.

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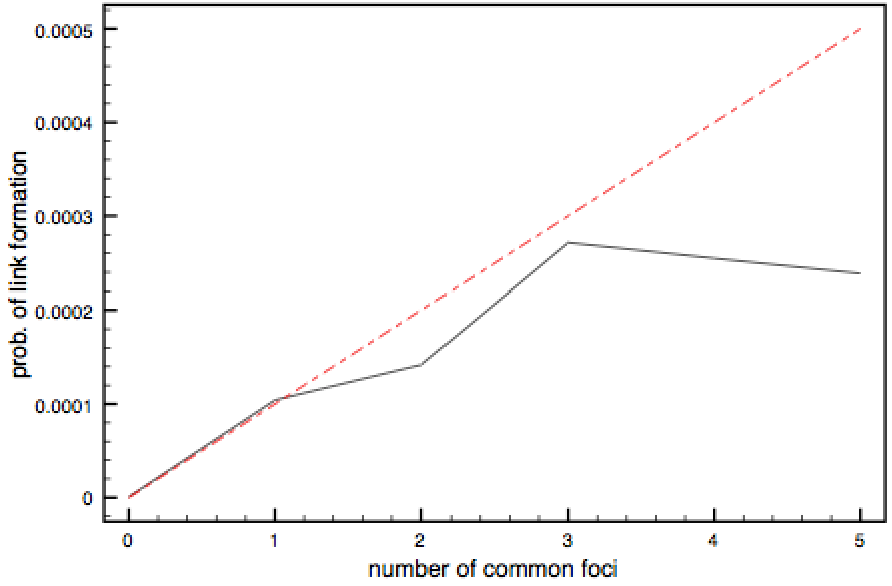
Example with $q = 0.20$:

Shared contacts k	1	2	3	5
$P(\text{new tie} \mid k)$	0.20	0.36	0.49	0.67

The first shared contact adds the most. Later shared contacts add less because many opportunities are redundant.

Focal Closure at Scale



Source: Easley & Kleinberg 2010

Setup. Two people who have **never emailed each other** but co-enroll in k classes during the same semester. What is the probability they exchange email within the term?

Result. Same qualitative shape as triadic closure: sharp rise from $k = 0$, then saturation. Sharing one class is already informative; 3+ classes produces a much higher closure rate.

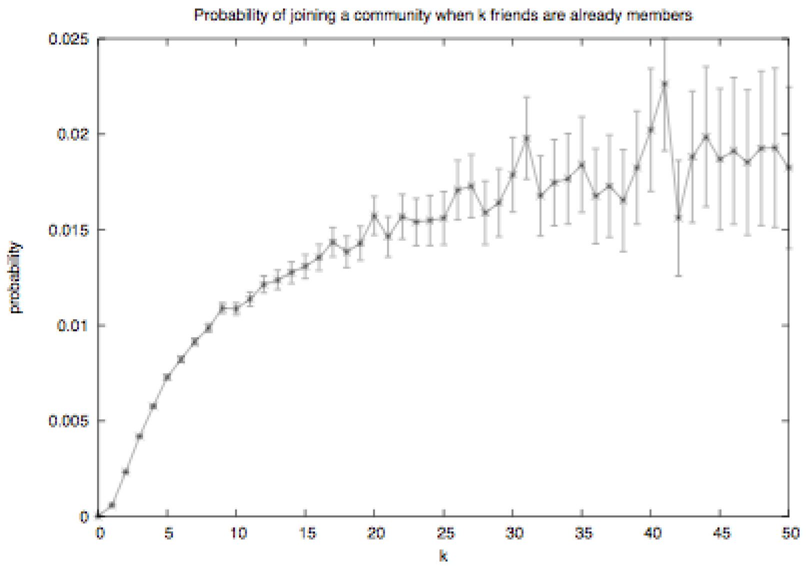
Why this matters.

- Focal closure runs **without any shared friend**. Strangers in the same context form ties through repeated exposure alone.
- This is the mechanism that turns "people share classes" into "people share friendships" — and it produces homophily-like patterns even when

there is **no preference for similarity at all**.

- It is L05's main answer to *Question 3* from the opening slide ("could shared context explain the clustering?"). Often, yes.

Membership Closure at Scale



Source: Easley & Kleinberg 2010

Setup. A person is **not yet a member** of a focus (course, club, group). How does the probability of joining depend on the number of existing members who are already their friends?

Result. Same saturating shape. One friend already in the focus is a strong predictor of joining; additional friends compound — but with diminishing returns.

The flipped direction.

- Triadic and focal closure close *person–person* ties.
- Membership closure closes a *person–focus* tie — your friends pull you **into** the contexts they already inhabit.
- This is how **social networks shape affiliations**, not just the other way around. Friendships and foci co-

Quick Check

A study finds that the probability of two coworkers becoming Facebook friends rises from 1% (no shared friends) to 20% (one shared friend) to 30% (two) to 33% (three or more).

1. Describe the shape of the curve.
2. Fit the independent-opportunity model: what value of q gives $P = 0.20$ at $k = 1$?
3. Does the model predict the observed $P = 0.30$ at $k = 2$? What does the mismatch tell you?

≡

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1. Steep initial rise from 1% to 20%, then much smaller gains (30%, 33%) — a **saturating** curve.

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2. At $k = 1$: $1 - (1 - q) = q$, so $q \approx 0.20$.

3. The model predicts $1 - (1 - 0.20)^2 = 0.36$ but we observe **0.30** — a shortfall. Shared friends are *correlated* (typically embedded in the same group), so they provide less independent opportunity than the model assumes. Diminishing returns are *steeper* than independence predicts.

Takeaway

Online data turns link formation into a measurable, time-stamped process. Empirical closure curves show saturation — the first shared contact matters most, and additional contacts are redundant rather than independent.

From Local Preferences to Global Segregation

> **Guiding Question:** Can weak local preferences generate strong global segregation?

Schelling's Threshold Model

Setup. Agents of two types are arranged on a grid (or network). Each agent i has a threshold $\tau \in [0, 1]$ — the minimum fraction of same-type neighbors it needs to be "satisfied".

Update rule.

1. Identify all *unsatisfied* agents (fewer than fraction τ of neighbors are same-type).
2. Each unsatisfied agent moves to a random vacant cell (or rewires a random tie in a network setting).
3. Repeat until no agent wants to move — or until a stable oscillation is reached.

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Schelling's insight (1971). Even τ as low as $1/3$ — "I just want a third of my neighbors to be like me" — produces **sharply segregated** global patterns.

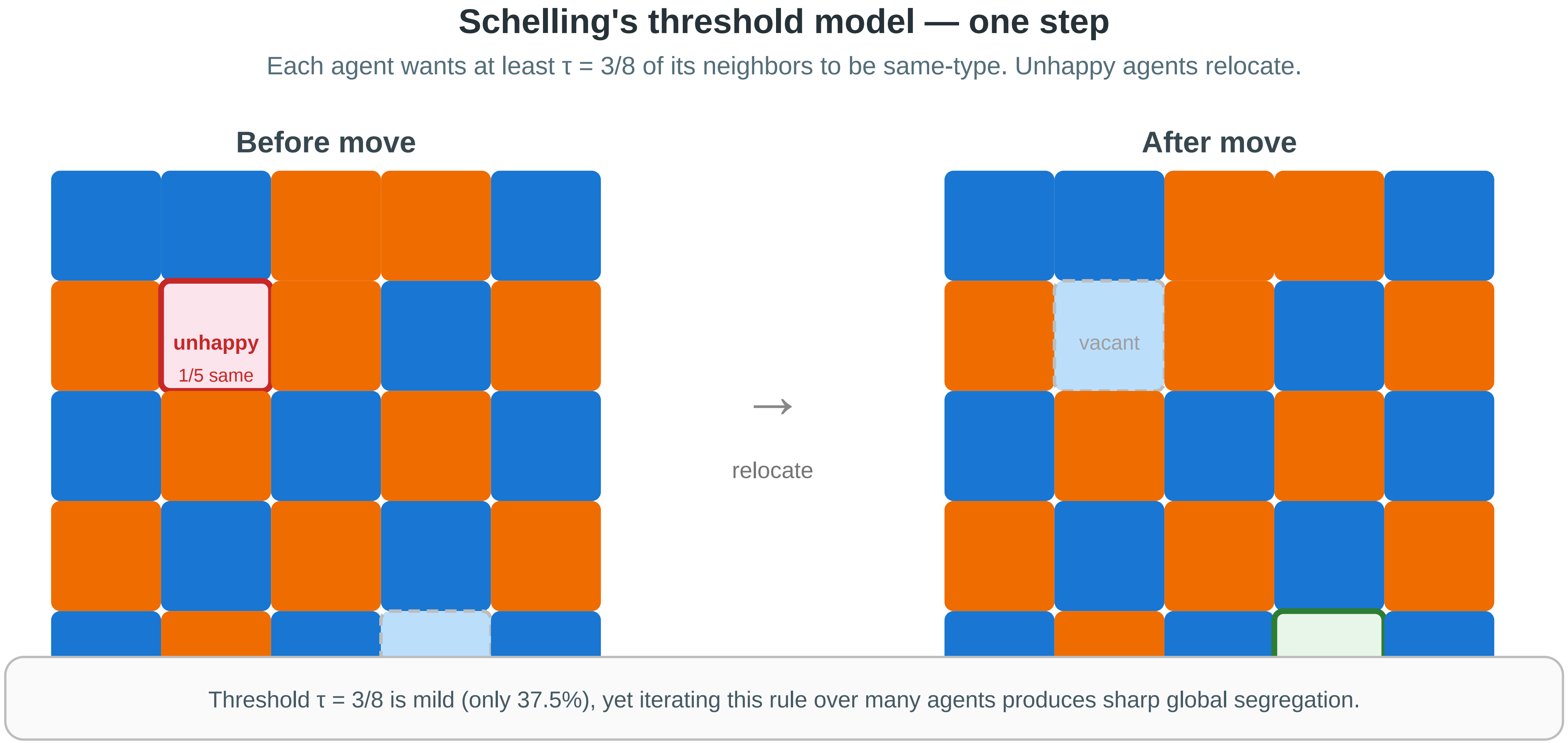
Schelling — A Local Rule

Let $\tau = 1/3$. An agent compares its local neighborhood to that threshold.

Same-type neighbors	Total neighbors	Same-type fraction	Status
2	8	0.25	Unsatisfied: move or rewire
3	8	0.375	Satisfied: stay
4	8	0.50	Satisfied: stay

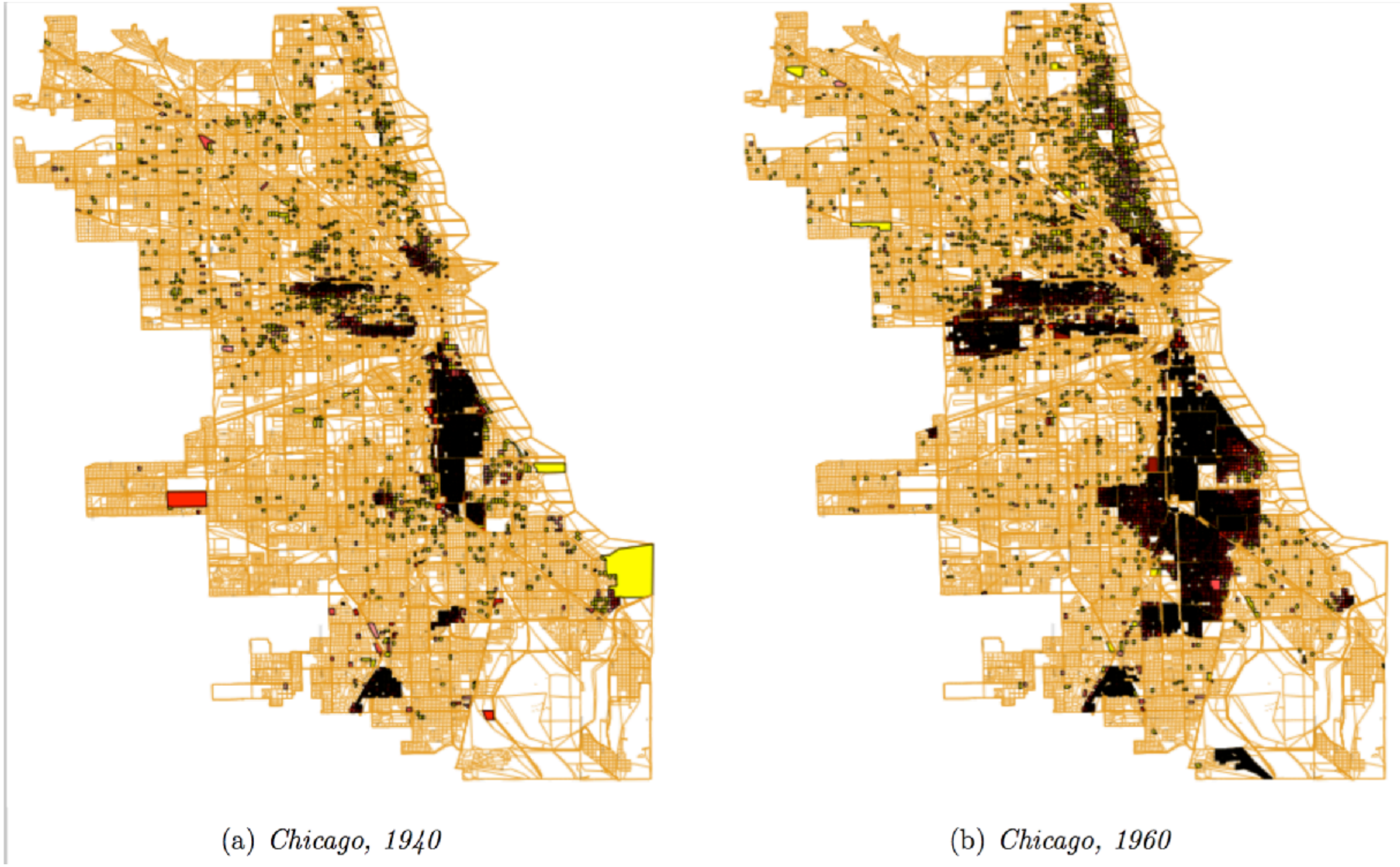
The rule is mild: the agent does **not** require a majority of similar neighbors. Strong segregation emerges because many mild local moves are repeated.

One Step of the Model



The blue agent at (1,1) has 1/5 same-type neighbors — below $\tau = 3/8$. It relocates to vacant cell (4,3), where 3/5 neighbors are same-type — satisfied. Repeat across all unsatisfied agents until equilibrium.

Empirical Motivation



Source: Easley & Kleinberg 2010

Residential segregation in Chicago by race/ethnicity. The sharp boundaries between neighborhoods are consistent with Schelling dynamics: each household's mild preference for "some" same-type neighbors, iterated over decades of moves, produces stark global segregation.

From Grids to Networks

In a network setting, Schelling's model becomes a **rewiring** process:

- "neighbors" are graph-adjacent nodes
- "moving" becomes dropping a tie to a dissimilar contact and forming a new tie to a similar contact
- the threshold τ controls how much same-type concentration triggers rewiring

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- the threshold τ controls how much same-type concentration triggers rewiring

Implication for recommendation algorithms. If a platform suggests contacts who are *similar* to your current contacts, it acts as an automated Schelling rewirer — accelerating homophily-driven segregation even beyond what individual preferences would produce alone.

Quick Check

Translate Schelling's spatial model into a network setting.

1. What plays the role of "neighbor" in a network rather than a grid?
2. What plays the role of "moving"?
3. A social media platform recommends friends similar to your existing friends. In Schelling terms, what is the platform doing — and would you expect it to amplify or weaken homophily?

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2. Moving: rewiring — dropping ties to dissimilar contacts and forming new ones with similar contacts.

3. The platform is an automated Schelling rewirer. It surfaces similar people — equivalent to offering each agent a move toward a higher same-type fraction. This *amplifies* homophily: the local rule ("recommend similar") is mild, but at machine speed it produces sharp global outcomes (filter bubbles, echo chambers).

Takeaway

> Global polarization can emerge without anyone explicitly aiming for it. Mild local preferences plus iterated local updates produce sharp large-scale structure — and recommendation algorithms now run this process at scale.

Wrap-Up — The Three Gaps

Across Lectures 03–05, the course has exposed three kinds of gap between what we *want to know* about a network and what the data *actually shows*:

Lecture	Gap type	Theoretical ideal	Practical limitation	Strategy
L03	Computational	True tie labels (MaxSTC)	NP-hard	Polynomial proxies (clustering, overlap)
L04	Computational	Optimal partition ($\max Q$)	NP-hard	Heuristics (GN, Louvain, Leiden)
L05	Causal	Mechanism (selection vs. socialization)	Not identifiable from snapshots	Longitudinal data, affiliation models

Every concept in the three lectures occupies one cell of this table.

Looking Ahead: Lecture 06

If signs (positive and negative) — not just connection strengths — matter, how do whole social structures stabilize or destabilize? Structural balance theory turns the question of friend-and-enemy relations into a graph-theoretic claim with measurable predictions.

Reading

Chapter 4 of Easley & Kleinberg (2010) covers homophily, selection, and socialization. McPherson, Smith-Lovin & Cook (2001), *Birds of a Feather*, is the definitive review of homophily research. Christakis & Fowler (2007) is the Framingham obesity study; Lyons (2011) provides the methodological critique.

Image Credits & References

Cinelli, Matteo and De Francisci Morales, Gianmarco and Galeazzi, Alessandro and Quattrociocchi, Walter and Starnini, Michele (2021). The echo chamber effect on social media. *Proceedings of the National Academy of Sciences*.

Easley, David and Kleinberg, Jon (2010). Networks, Crowds, and Markets: Reasoning about a Highly Connected World. *Cambridge University Press*. https://www.cs.cornell.edu/home/kleinber/networks-book/

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